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Towards linking lab and field lifetimes of perovskite solar cells

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Metal halide perovskite solar cells (PSCs) represent a promising low-cost thin-film photovoltaic technology, with unprecedented power conversion efficiencies obtained for both single-junction and tandem applications[1–8]. To push PSCs towards commercialization, it is critical, albeit challenging, to understand device reliability under real-world outdoor conditions where multiple stress factors (for example, light, heat and humidity) coexist, generating complicated degradation behaviours[9–13]. To quickly guide PSC development, it is necessary to identify accelerated indoor testing protocols that can correlate specific stressors with observed degradation modes in fielded devices. Here we use a state-of-the-art positive-intrinsic-negative (p–i–n) PSC stack (with power conversion efficiencies of up to approximately 25.5%) to show that indoor accelerated stability tests can predict our six-month outdoor ageing tests. Device degradation rates under illumination and at elevated temperatures are most instructive for understanding outdoor device reliability. We also find that the indium tin oxide/self-assembled monolayer-based hole transport layer/perovskite interface most strongly affects our device operation stability. Improving the ion-blocking properties of the self-assembled monolayer hole transport layer increases averaged device operational stability at 50 °C–85 °C by a factor of about 2.8, reaching over 1,000 h at 85 °C and to near 8,200 h at 50 °C, with a projected 20% degradation, which is among the best to date for high-efficiency p–i–n PSCs[14–17].

Metal halide perovskite solar cells (PSCs) or photovoltaics are considered technologically important to enable low-cost, high-efficiency, large-scale (terawatt-level) applications[18]. Single-junction PSCs have already reached power conversion efficiencies (PCEs) beyond those of other polycrystalline thin-film photovoltaic technologies[1]. Understanding and improving the long-term operational stability of PSCs to ensure reliability in outdoor field conditions has become a critical and challenging task to transition PSCs from lab to market[10–13,19–21]. When deployed for terrestrial applications, solar cells must endure a set of extreme and harsh conditions (often with variable combinations of ever-changing stress factors) during their service lifetime. For most research groups, stability tests are conducted indoors with a few fixed stressing conditions. These tests provide insight into the stability of cell operation as well as critical material and interface requirements for stable devices. Ultimately, however, it is important to understand, quantitatively, the correlation between indoor tests and outdoor field operation, which requires assessing and identifying key degradation pathways with associated failure modes and their relationships to the device components.

To reach the stability and reliability targets for commercialization, it is necessary to first establish protocols for testing and reporting so that the improvements in stability from different groups can be easily validated, compared and understood for further advances. Consensus has largely been achieved for test conditions and reporting, with the 2020 International Summit on Organic Photovoltaic Stability (ISOS) article providing recommended best practices for PSC researchers to use[9]. Among various stability testing protocols, researchers tend to focus on low-temperature operational stability under light, sometimes in combination with dark stability tests at elevated temperatures (Extended Data Fig. 1 and Supplementary Table 1). Although most recent studies have followed ISOS protocols, given the wide range of ISOS testing conditions, there is a lack of data that directly correlates ISOS tests particularly across indoor and outdoor conditions. Thus, it is often still challenging to compare different studies and discern their relevance to achieving the reliability required for commercialization[22]. It is critical to identify a subset of ISOS protocols, which can be used collectively to quantitatively link indoor with outdoor device degradation behaviour and predict outdoor durability, to accelerate the development of PSCs for practical use.

In this study, we focused on cell-level stability—for both indoor and outdoor operations—based on inverted (or p–i–n) PSCs, with lab PCE up to about 25.5% and certified stabilized PCE reaching 24.3%. We found that these devices exhibited faster degradation under operation at elevated temperature (T) from 25 °C to 85 °C, from which we obtained an apparent activation energy (E_a) of approximately 0.59 eV for the degradation rates. The outdoor ageing behaviour of well-packaged devices also varied with temperature as the seasons changed. This can

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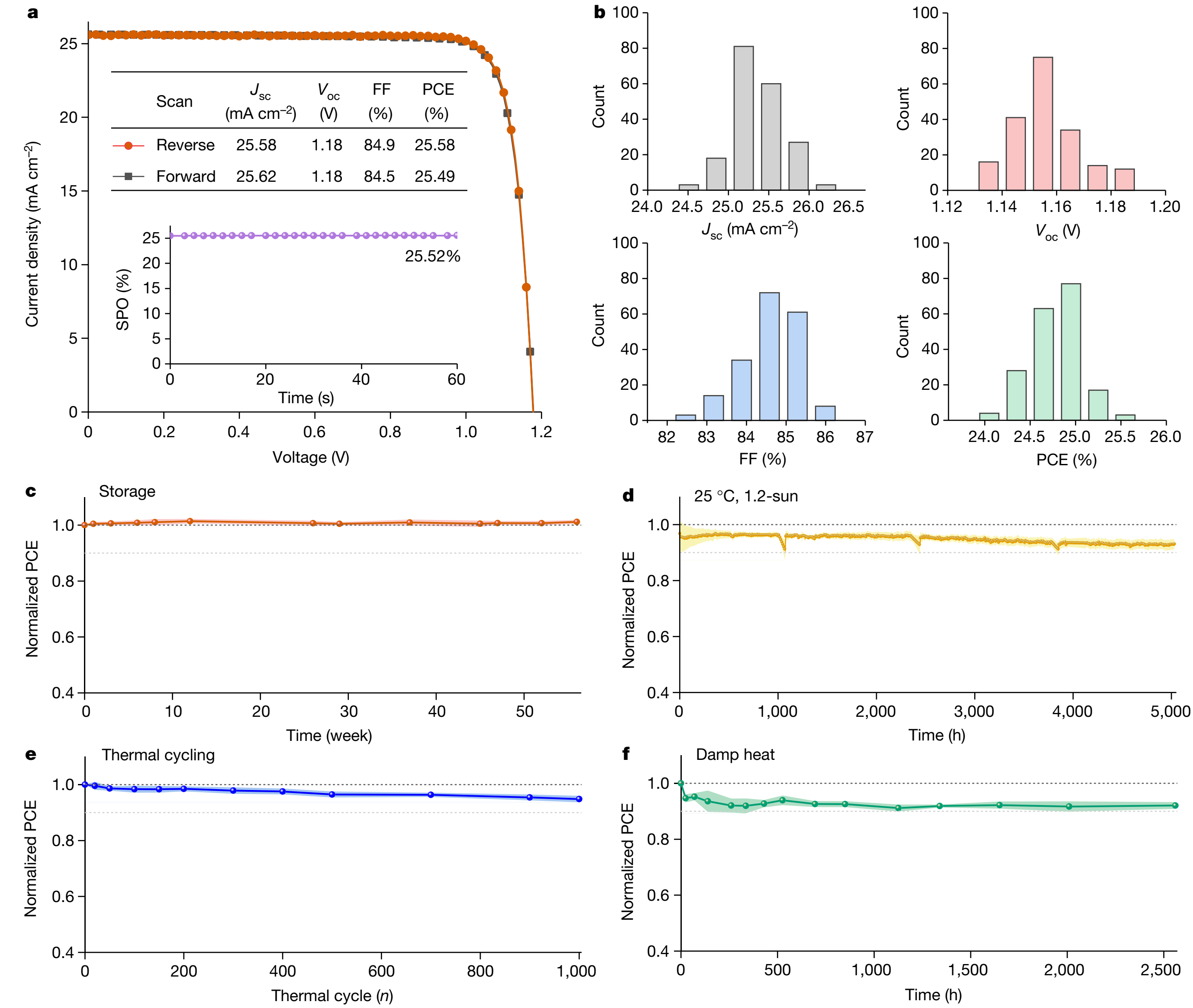

Fig. 1 | Device characteristics and stability. a, *J*–*V* curves of a typical p–i–n PSC (glass/ITO/MeO-2PACz/$Rb_{0.05}Cs_{0.05}MA_{0.05}FA_{0.85}Pb(I_{0.95}Br_{0.05})_3$/$C_{60}$/$SnO_2$/Ag) and the corresponding SPO PCE. **b**, Statistical comparison of photovoltaic parameters from 192 devices. FF, fill factor. **c**,**d**, Device efficiency evolution for unpackaged PSCs under storage in dark in N_2 (ISOS-D-1I) (**c**) and operation under continuous 1.2-sun at 25 °C in N_2 (ISOS-L-1I) (**d**). **e**,**f**, Device efficiency evolution for packaged PSCs under repeated thermal cycling (−40 °C to 85 °C) in the dark in air (ISOS-T-3) (**e**) and under damp heat testing at 85 °C and 85% relative humidity in the dark in air (ISOS-D-3) (**f**). We used 5–12 individual devices for each stability test in **c**–**f** to obtain the average degradation behaviour, with the error bars representing the standard deviations.

be modelled by the *T*-dependent degradation rates from indoor accelerated stability tests under light and heat. Conducting indoor operational testing under illumination and high temperature (for example, 85 °C, ISOS-L-2) allowed us to rapidly identify and mitigate a critical instability mechanism causing device instability associated with the indium tin oxide (ITO)/hole transport layer (HTL)/perovskite interface in our PSCs. A simple modification of the HTL-improved averaged device stability by a factor of about 2.8 at 50–85 °C under 1.2-sun, reaching a projected T80 (time for a solar cell to degrade to 80% of its maximum efficiency) of over 1,000 h at 85 °C and to near 8,200 h at 50 °C, which is among the best to date for high-efficiency p–i–n PSCs.

Device property and indoor stress tests

In this study, we focused on an inverted device stack adapted from our recent report[2]. A typical device consisted of glass/ITO/(2-(3,6-Dimethoxy-9H-carbazol-9-yl)ethyl)phosphonic acid (MeO-2PACz)/$Rb_{0.05}Cs_{0.05}MA_{0.05}FA_{0.85}Pb(I_{0.95}Br_{0.05})_3$/$C_{60}$/tin oxide ($SnO_2$)/silver (Ag), where MeO-2PACz denotes a phosphonic acid functionalized carbazole self-assembled monolayer (SAM) that behaves as a hole contact. Representative current density–voltage (*J*–*V*) characteristics are shown in Fig. 1a, with PCEs of 25.58% and 25.49% from reverse and forward scans, respectively. The detailed photovoltaic parameters are given in the inset in Fig. 1a. The corresponding stabilized power output (SPO) PCE reached 25.52%, representing the state-of-the-art for p–i–n PSCs[14–17]. Figure 1b shows the statistical distribution of all photovoltaic parameters—short-circuit current density (J_{sc}), open-circuit voltage (V_{oc}), fill factor and PCE—from 192 devices, indicating good device reproducibility. This reproducibility is critical for obtaining reliable and consistent data from the different stability tests undertaken in this study. One typical device was measured by an accredited photovoltaics laboratory, obtaining a certified stabilized PCE of 24.3%

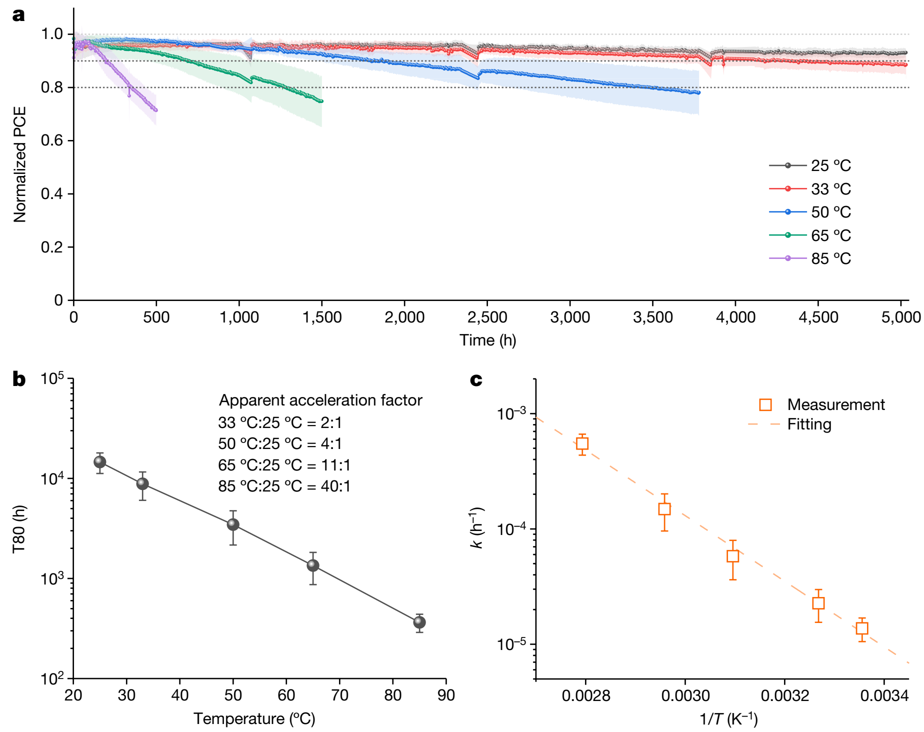

Fig. 2 | Temperature-dependent stability study. a, Averaged operational stability of unencapsulated p–i–n PSCs under continuous 1.2-sun in N_2 at five different temperatures (25°–85 °C, as indicated). **b**,**c**, Comparison of the temperature-dependent T80 values (**b**) and the corresponding hourly degradation rates (k = 1/T80) (**c**). The error bars represent the standard deviations from 8–17 individual devices for each temperature.

(Extended Data Fig. 2), which is the highest among the results in the literature to date[2].

To examine device efficiency evolution, we first applied several ISOS tests, including storage stability, operational stability at room temperature (25 °C), damp heat (85 °C/85% relative humidity)[9,10], and thermal cycling (−40 °C to 85 °C)[15,23,24]. For the storage stability test (dark, N_2; ISOS-D-1I), the devices showed no degradation after 56 weeks (over one year) (Fig. 1c). For the operational stability test at room temperature (25 °C, 1.2-sun, N_2; ISOS-L-1I), the devices retained more than 93% of their maximum PCEs after approximately 5,030 h of continuous operation (Fig. 1d).

Thermal cycling and damp heat tests have recently attracted increasing attention as they mirror common International Electrotechnical Commission tests used for outdoor conditions and commercialization[15,17,25–27]. In these two tests, we packaged the devices using a desiccant impregnated polyisobutylene (PIB) edge seal between two glass sheets (Extended Data Fig. 3), as detailed previously[28]. For the thermal cycling tests (−40 °C to 85 °C, dark; ISOS-T-3), our devices showed an average of approximately 5% degradation after 1,000 thermal cycles, with multiple cells displaying less than 3% degradation (Fig. 1e). Note that this well exceeds the 200 thermal cycles normally conducted for this testing protocol[10,29]. For the damp heat test (85 °C/85% relative humidity, dark; ISOS-D-3), our devices showed less than 8% degradation after about 2,560 h, which is also among the best reported to date.

Note that the stability results of the storage, room-temperature operation, thermal cycling and damp heat tests can, separately, address different stressors (for example, light, heat and humidity), testing the robustness of our p–i–n PSCs and device packaging. In real-world operation, however, these individual stress factors will act simultaneously to impact device operation and produce degradation. For example, when combined, light and heat substantially accelerate degradation and/or cause new degradation pathways that were absent or occurred at slower rates when tested separately[13,19]. Identifying and understanding the reliability limiting mechanism requires research efforts that iterate between outdoor and indoor testing of devices operating under a combination of stress conditions. It is critical to establish these connections, especially those that can enable a mechanistic understanding of real-world outdoor operation.

Temperature effect and outdoor ageing

We further examined the device stability under 1.2-sun illumination with elevated temperatures ranging from 25 °C to 85 °C to understand the impact of temperature in combination with illumination. These tests were conducted in N_2 consistent with our damp heat and thermal cycling tests (Fig. 1e, f), which demonstrate that the potential impact of moisture in ambient air can be minimized by reliable device packaging. Figure 2a shows the results of the temperature-dependent operational stability evolution. The nominally identical cells degraded faster on average with increasing device operation temperature, with the T80 changing from approximately 14,580 h at 25 °C to approximately 360 h at 85 °C. The T80 values were averaged from 8–17 independent devices for each temperature, and for the 25 °C and 33 °C tests the T80 values were extrapolated from 5,030 h of measurements. Note that J–V measurements were conducted to track device stability and the reverse-scan PCEs were used for analysis as the reverse- and forward-scan PCE evolutions are almost the same (Extended Data Fig. 4). For the samples examined in this study, the devices aged at 85 °C exhibited 40-fold faster degradation than the devices aged at 25 °C. Figure 2b contains the corresponding average T80 values at five different measurement temperatures along with the apparent acceleration factors relative to the 25 °C results. Figure 2c compares the hourly degradation rates (1/T80) at different temperatures. From this plot, we can estimate an apparent E_a of approximately 0.59 eV to reasonably describe the rate-limiting degradation process across the temperature range from 25 °C to 85 °C. This thermally activated degradation behaviour is comparable to a recent study that showed a single E_a value of about 0.43 to 0.49 eV for all-inorganic $CsPbI_3$-based PSCs[30].

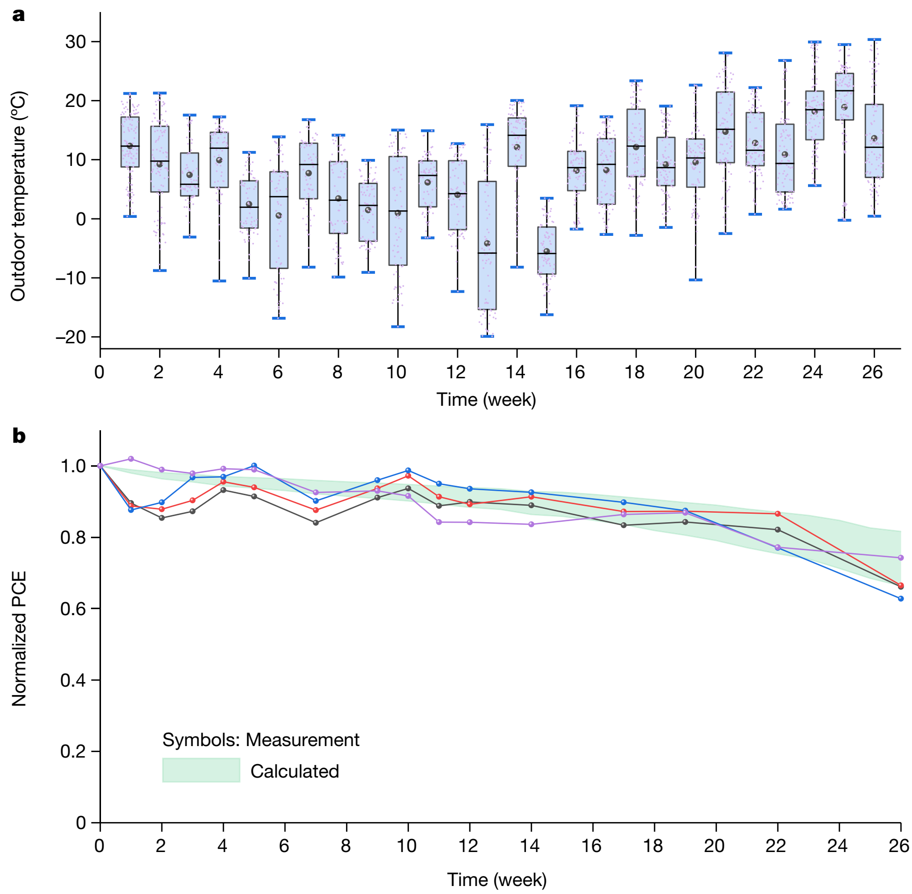

Fig. 3 | Outdoor ageing test. a, Outdoor ambient temperature trend. The median value, mean value, maximum and minimum values, and 25%–75% region of data are represented by the horizontal line (across the box), circle, top and bottom bars, and box, respectively. **b**, Device efficiency evolution for glass-PIB-packaged PSCs biased near maximum power point under outdoor ageing conditions. Four individual packaged devices were used in this test. The light green band shows the calculated degradation range based on the temperature-dependent hourly degradation rates derived from Fig. 2c. Note that the temperature inside a packaged PSC can reach up to 40°–50 °C above the ambient temperature; the lower and upper bounds of the light green band correspond to the ambient $T + 50$ °C and $T + 40$ °C, respectively.

We further investigated device stability under outdoor ageing conditions (ISOS-O-1). These devices were packaged with a PIB edge seal between glass sheets[28]. Because a higher temperature is more detrimental to device operation under illumination (Fig. 2), we tracked the ambient temperature variation during the outdoor ageing test. The weekly temperature trend during the outdoor stability test period is shown in Fig. 3a. Note that we were not able to add sensors inside the packaged devices to directly monitor the actual device temperature variation in this study. As an approximation, we found that the temperature inside a closed glass container under direct sunlight can increase by up to 40–50 °C above the ambient temperature. This is consistent with early studies that show solar cells experience a wide range of temperature fluctuations during outdoor operation[31–33]. Note that we periodically took the outdoor devices back indoors to measure the device PCE under a solar simulator with fixed conditions (AM1.5 G, 1 sun) to monitor the device degradation behaviour. This makes it easier to compare the long-term performance. Monitoring device degradation under real daylight with the same precision is difficult because of the changing environment (for example, temperature and irradiance), considering that varying temperature and irradiance affects photovoltaic characteristics (Extended Data Fig. 5) and in situ monitoring data would be substantially more complex to evaluate the link of the device degradation under outdoor condition with those indoors[34].

We used the temperature-dependent hourly degradation rates and outdoor testing temperatures to model the extent of device degradation during the outdoor operation; the analysis details are given in the Methods section. The results are shown in the light green band of Fig. 3b, with the lower and upper bounds corresponding to the ambient $T + 50$ °C and $T + 40$ °C, respectively. For the outdoor stability test, we also included calcium samples inside the packaged devices as a sensitive probe to monitor potential moisture ingress[28]. For the devices shown in Fig. 3b, we did not see any obvious moisture ingress (Extended Data Fig. 3). This ensured that the devices evaluated were not subject to further moisture-related stress, which would complicate our analysis of device degradation. Note that we found degradation/recovery associated with diurnal (day–night) cycles is not critical for our device stack (Extended Data Fig. 6). The direct ultraviolet (UV) exposure tests also showed a relatively small impact of UV stress on the device degradation in this study (Extended Data Fig. 7). Figure 3b shows the calculated degradation behaviour, which mimics the measured outdoor device degradation results. The devices maintained approximately 66–75% of initial PCEs after 26 weeks under outdoor conditions. This indicates that the combination of high temperature with illumination is the most critical stressing condition and, in the absence of moisture ingress for well-packaged devices, it can be used to quantitatively predict the cell performance under outdoor operational conditions for the test period.

Interface modification and stability

The faster device degradation at a higher temperature under illumination can be attributed to multiple factors, such as chemical reactions at interfaces and ion migration[35,36]. Over the past several years, interface

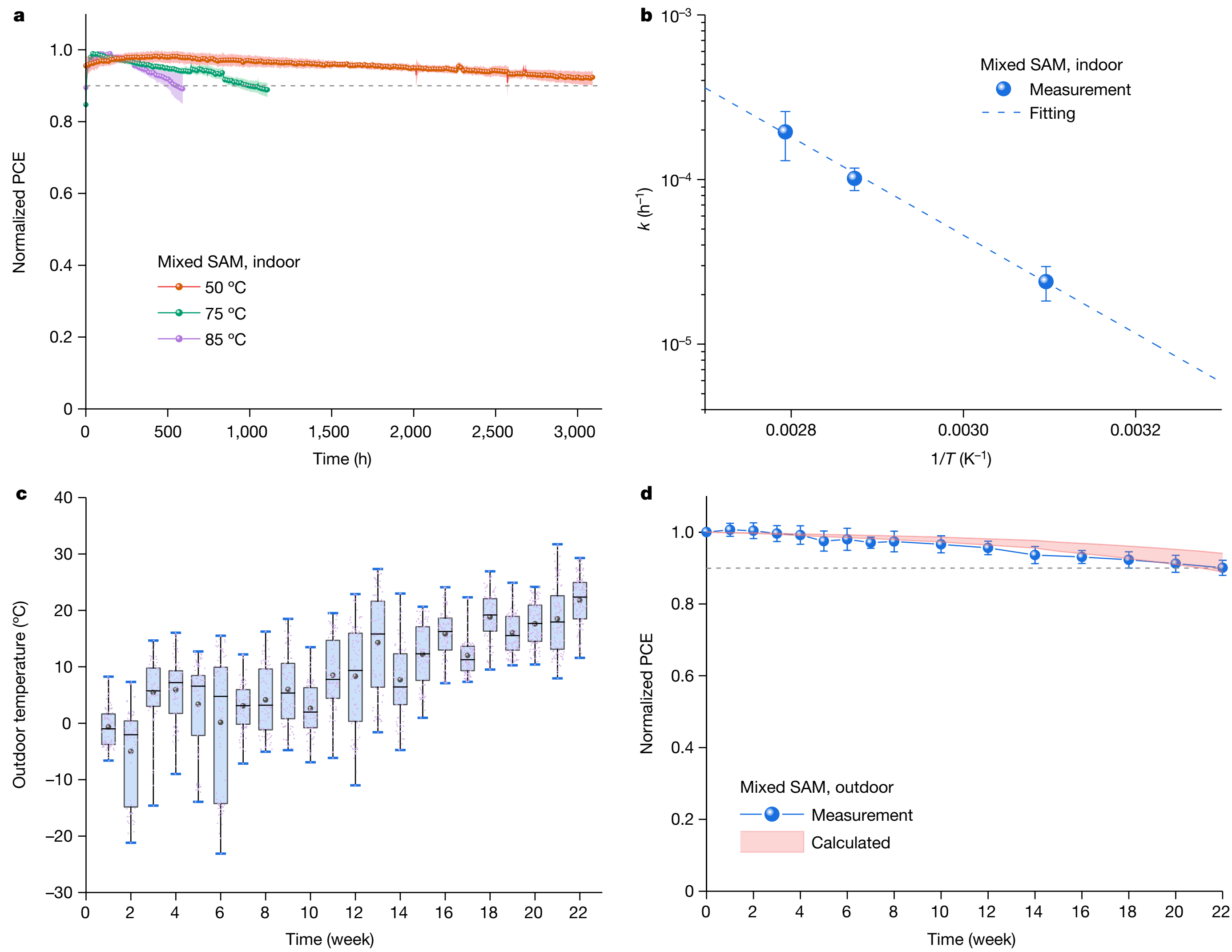

Fig. 4 | Outdoor and indoor stability tests based on mixed SAM HTL. **a**, Averaged operational stability of unencapsulated p–i–n PSCs under continuous 1.2-sun in N_2 at three different temperatures (50 °C, 75 °C, 85 °C, as indicated). **b**, Hourly degradation rates derived from the corresponding T80 values at different temperatures. The error bars represent the standard deviations from 10 to 20 individual devices for each temperature. **c**, Outdoor ambient temperature trend. The median value, mean value, maximum and minimum values, and 25–75% region of data are represented by the horizontal line (across the box), circle, top and bottom bars, and box, respectively. **d**, Normalized PCE evolution of packaged devices under outdoor ageing conditions. The pink band shows the calculated trend. The error bars represent the standard deviations from 14 individual packaged devices.

engineering has shown great promise in improving PSC efficiency and stability[2,15–17]. As we showed previously, the perovskite/electron transport layer (ETL) interface in our device stack is optimized with the reactive 3-(aminomethyl)pyridine (3-APy) surface treatment, which creates a surface field to facilitate charge separation and minimize charge recombination on the ETL side of the device stack[2]. Additionally, we have shown that ITO/perovskite interfaces are thermally and electrochemically reactive[35]. Thus, we suspect that the bottom interface region (ITO/HTL/perovskite) is likely less than ideal, limiting our device stability under illumination at high temperatures. Density functional theory (DFT) calculations and electrochemical measurements showed that the HTL based on mixed SAMs of MeO-2PACz and Me-4PACz (4-(3,6-dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid) has enhanced ion-blocking properties compared to MeO-2PACz (HTL in Figs. 1–3), leading to higher interfacial stability at elevated temperatures (Extended Data Fig. 8 and Methods).

We examined the indoor and outdoor operational stabilities of p–i–n PSCs based on a MeO-2PACz and Me-4PACz mixed SAM. As shown in Fig. 4a, the averaged T80 reached over 1,000 h at 85 °C (extrapolated from approximately 600 h of measurement) and further increased to near 8,200 h at 50 °C (extrapolated from approximately 3,100 h of measurement), corresponding to a factor of approximately 2.8 improvement with a similar activation energy (approximately 0.59 eV) (Fig. 4b) in comparison to the devices based on MeO-2PACz (Fig. 2). The activation energy agreement strongly suggests the rate-limiting degradation reactions are the same mechanism in both types of devices. The average T80 values were obtained based on 10–20 individual devices for each temperature. The packaged devices were aged under outdoor conditions for 22 weeks. Fourteen packaged devices maintained an average of 90.1% of initial device PCEs (Fig. 4c,d). Note that reverse *J*–*V* scan PCEs were used to track the stability considering the comparable results, with the same trends observed using reverse/forward/maximum power point tracking values (Extended Data Fig. 9). The degradation trend is consistent with the calculated ones based on temperature-dependent hourly degradation rates and outdoor ageing temperature. Our second set of devices in this study, with improved HTL and stability, verified again that the combination of high temperature with illumination is the most critical condition to determine a cell's degradation under outdoor conditions for the test period. These results underscore the importance of identifying and improving the weakest components of PSCs for stable device operation at high temperatures; this is critical to enabling PSC development for outdoor applications.

In conclusion, in this study, we demonstrate a large set of stability assessments using a state-of-the-art p–i–n PSC platform consisting of multiple ISOS stability testing protocols. We identify that conducting high-temperature (for example, 85 °C) operational stability testing under light illumination is well correlated to outdoor operational cell testing. This indicates that a combination of high temperature and illumination is the most critical combination of stressors for understanding outdoor operation of well-packaged PSCs. Our results also show that the ITO/SAM HTL/perovskite interface is critical to enhancing device operation stability at high temperatures and it should receive more attention in future research on p–i–n PSC development for stable and efficient outdoor operation. Assuming no change in the primary degradation mechanism, testing at a higher temperature under illumination can also accelerate the development cycle of high-performance PSCs to advance the perovskite photovoltaic industry. Although other cell-level ISOS stability test protocols are still valuable for providing comparisons to guide PSC development, particularly in moving from cells to modules, more stringent tests combining multiple ISOS protocols and associated stressors should be considered to improve the understanding of outdoor device operation.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-023-06610-7.

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Methods

Materials

Rubidium iodide (RbI, 99.99%), caesium iodide (CsI, 99.99%) and lead iodide (PbI_2, 99.999%) were purchased from Sigma-Aldrich. Organic salts of formamidinium iodide (FAI) and methylammonium bromide (MABr) were purchased from Greatcell Solar Materials; lead bromide ($PbBr_2$), (2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl)phosphonic acid (MeO-2PACz) and (4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid (Me-4PACz) were purchased from Tokyo Chemical Industry. Dimethylformamide (DMF, 99.8%, anhydrous), dimethyl sulfoxide (DMSO, 99.9%, anhydrous), chlorobenzene (99.9%, anhydrous) and toluene (99.9%, anhydrous) were purchased from Sigma-Aldrich. C_{60} was purchased from Lumtec. All chemicals were used directly as received.

Device fabrication

The device fabrication details are similar to our previous report[2]. In brief, the patterned conducting oxide glass indium tin oxide (ITO) substrates were washed with acetone and isopropanol for 15 min each. After 15 min of ultraviolet ozone treatment for the substrates, 0.5 mg ml^{-1} of MeO-2PACz or a mixed self-assembled monolayer (SAM) ethanol solution of MeO-2PACz:Me-4PACz (v:v = 4:1 to 1:1) was spin-coated onto the substrates at 3,000 r.p.m. for 30 s, processed in a nitrogen glovebox. Then the substrates underwent 100 °C annealing for 10 min. The perovskite composition is $Rb_{0.05}Cs_{0.05}MA_{0.05}FA_{0.85}Pb(I_{0.95}Br_{0.05})_3$. The perovskite precursor was prepared by using a stoichiometric molar ratio of CsI, RbI, FAI, PbI_2, $PbBr_2$ and MABr in a mixed solvent of DMF:DMSO (v:v = 4:1). An antisolvent method was used to make the intermediate perovskite film. During a typical perovskite coating, the substrate was first spinning at 1,000 r.p.m. for 10 s and then at 3,000 r.p.m. for 40 s. After 15 s into the second stage, 150 µl chlorobenzene antisolvent was dropped on top of the spinning substrates. The perovskite sample was subsequently annealed at 100 °C for 10 min. We used 0.1 mM of 3-APy chemical as a posttreatment at 5,000 r.p.m. for 30 s, followed by annealing at 70 °C for 5 min. Afterwards, samples were transferred to an Angstrom evaporator for coating of 25 nm C_{60}, followed by coating 30 nm SnO_2 by atomic layer deposition and finally evaporated with 100 nm Ag or Au as a metal electrode. The atomic layer deposition SnO_2 process is detailed in our previous report[37]. The device area by evaporation was 0.122 cm^2. Unless otherwise stated, the devices were masked with metal aperture masks (0.059 cm^2) during the efficiency measurements.

Current density–voltage measurement

The *J*–*V* characteristics of unpackaged devices were measured in an N_2-filled glovebox using a Newport Oriel Sol3A class AAA solar simulator with a xenon lamp and the power of the light was calibrated to 1-sun AM1.5 G illumination by a Si reference cell with a KG2 filter. All devices were measured using a Keithley 2400 source meter under a sweep mode of reverse scan (from 1.2 V to −0.1 V) and forward scan (from −0.1 V to 1.2 V). The efficiencies of packaged devices—for damp heat, thermal cycling and outdoor tests—were measured in air under simulated 1-sun AM1.5 G illumination (Oriel Sol3A class AAA solar simulator, Newport Corporation).

Outdoor package procedures

The outdoor package details followed instructions for the 'flip-chip' package from a previous report[28]. All packaging steps were completed inside a nitrogen glovebox. Briefly, the 1 in (ref. 2) cell substrate was press-contacted onto a 3 in × 3 in glass substrate with deposited metal leads via indium pellets to form an electrical contact. Desiccant impregnated polyisobutylene tape (PIB; HB Fuller | Kommerling Helioseal PVS-101; size: 0.7 mm (thickness) × 10 mm (width)) was used as the package edge seal and was cut to line the edges of a 2 in × 2 in piece of 1.1-mm-thick Eagle glass. This coverglass piece with the PIB tape was set on a 1 in metal plate already heated on a hotplate at 135 °C until equilibration (approximately 15 min). The 3 in × 3 in side of the package with the contacted cell substrate was carefully flipped over onto the PIB/2 in × 2 in coverglass piece and 1 in metal plate. Then the entire package stack was moved into a pneumatic press and pressed for 2 min at 340 N cm^{-2} (500 lbf in^{-2}) of PIB tape area. A small card of Ca evaporated on polyethylene terephthalate (PET) was encased inside the package as an indicator of the package integrity (upon air/moisture exposure, the Ca will turn clear). To provide a static load while outside, an edge card providing a 510 Ω load on each desired cell was clipped onto the package to contact the leads. The card was removed when measuring device *J*–*V* characteristics. For further details, including glass cleaning and treatment procedures, metal deposition, edge card connector design and so on, please refer to ref. 28.

Thermal cycling testing

The packaged devices were placed in the thermal cycling test chamber (ESPEC Corp.), and underwent temperature cycling from −40 °C (10 min dwell) to 85 °C (10 min dwell), at a ramp rate of 100 °C per hour. The devices were taken out the chamber after 20, 50, 100, 200, 300, 400, 500, 700, 900 and 1,000 cycles, to take the *J*–*V* measurements.

Damp heat measurement

The damp heat test was conducted by putting packaged devices at 85 °C/85% relative humidity in the environment test chamber (large Blue M). The devices were taken out for *J*–*V* measurement after cooling down to room temperature.

Stability parameter analyser measurement

For long-term stability measurements, the solar cells were loaded into a custom-built stability measuring system, dubbed the stability parameter analyser; the setups were the same as in our previous report[38]. In summary, the setup consists of cooling/heating tubes to keep the cell housing at a specified temperature (as needed), a flow chamber to control the atmosphere/environment of the cells, and electrical housing and electronics to switch between devices, measure *J*–*V* curves, and holds the devices under resistive load, with a white light-emitting diode-based light source to provide constant illumination (Extended Data Fig. 10). In this study, devices were kept in a N_2 environment at different temperatures (25 °C, 33 °C, 50 °C, 65 °C, 75 °C and 85 °C) underneath a white light-emitting diode at 1.2-sun and held under a resistive load of 510 Ω (placing the cells close to maximum power point). Every 30 min, the system removed the resistive load and took a *J*–*V* scan using a Keithley 2450 source-measure unit. *J*–*V* curves were then analysed to extract relevant photovoltaic figures of merit.

Outdoor measurements and data analysis

For the packaged devices for outdoor testing, the devices were periodically taken back indoors, and measured in air under simulated 1-sun AM1.5 G illumination (Oriel Sol3A class AAA solar simulator, Newport Corporation). The outdoor temperatures were collected in hourly intervals during daylight and aggregated in weeks shown in Fig. 3a. For the calculation results shown in Fig. 3b, we used the following steps. The different degradation rates measured at different temperatures (symbols in Fig. 2c) can be fitted to an apparent exponential decay function with a single activation energy (the dashed line in Fig. 2c). From the fitted line, we can determine hourly degradation rates for a given temperature within the temperature range. We then use the temperature-dependent hourly degradation rate along with the estimated device temperatures and corresponding durations under sunlight to calculate the accumulated degradation for a given week when the devices were aged in the outdoor conditions. These weekly degradation rates are then used to calculate the total anticipated degradation as a function of the ageing duration in weeks.

Electrochemical measurement

Electrochemical measurements were taken with a CH Instruments Inc. CH660E potentiostat in an 80-ml cell filled with deionized water and methylammonium chloride (0.2 M). The counter and reference electrodes were Pt foil and Ag|AgCl (purchased from CH Instruments Inc.), respectively. Under these conditions, we observed bare ITO cathodic reduction when held at potentials lees than −1 V, and no reaction at more positive potentials. Methylammonium is a weak acid and greater than 99% remains protonated in solution, so we assume it is deprotonated upon reaching the ITO interface. We did not observe an increase in this threshold for ITO/SAM HTL samples, suggesting the same reaction as bare ITO.

Density functional theory calculation

For the model construction, we built an $FAPbI_3$ perovskite slab with six PbI_2 atomic layers, which was terminated by an FAI atomic layer on the surface with the vacuum region set to be more than 20 Å. The Me-4PACz and MeO-2PACz molecules were placed on the surface to be absorbed and interact with the surface. We employed the CP2K package to relax the single H_2O, Me-4PACz, and MeO-2PACz molecules and the $FAPbI_3$ surfaces with Me-4PACz and MeO-2PACz absorbed, respectively; this was carried out at the Perdew–Burke–Ernzerhof level[39,40]. The double-ξ valence polarization basis set, along with the norm-conserving Goedecker-Teter-Hutter (GTH) pseudopotential, were chosen in the calculation[41–43]. An energy cutoff value of 500 Rydberg was set, and only the Gamma (Γ) point (1 × 1 × 1) was considered. Grimme's D3 correction was included due to the existence of van der Waals interactions[44,45]. The static single-point energy calculations were then implemented with a convergence criterion of 1×10^{-7} Ha to obtain the total energy of these molecules and surface configurations. Based on the total energies calculated as such, the binding energy between Me-4PACz/MeO-2PACz and H_2O molecules can be expressed as:

$$E_b = E_{H_2O} + E_{mol} - E_{whole} \quad (1)$$

where E_b denotes the binding energy, E_{H_2O} denotes the total energy of the H_2O molecule, E_{mol} denotes the total energy of the single Me-4PACz/MeO-2PACz molecule and E_{whole} denotes the total energy of the Me-4PACz/MeO-2PACz molecule binding with H_2O. Similarly, the binding energy of the Me-4PACz/MeO-2PACz molecule interacting with the $FAPbI_3$ surface can be calculated as:

$$E_b = E_{surface} + E_{mol} - E_{whole} \quad (2)$$

where E_b denotes the binding energy, $E_{surface}$ denotes the total energy of the $FAPbI_3$ surface without the interacting molecule, E_{mol} denotes the total energy of the single Me-4PACz/MeO-2PACz molecule and E_{whole} denotes the total energy of the $FAPbI_3$ surface interacting with the Me-4PACz/MeO-2PACz molecule.

Simulated diurnal cycle dependence study

A simulated diurnal cycle test—with 12-hour light on and 12-hour light off cycles—was conducted with 23 cycles for approximately 550 h. The results of two typical devices from this test are shown in Extended Data Fig. 6. A transient behaviour was shown during the initial several cycles that the PCE dropped slightly (approximately 3–4%) in a few hours when light is switched on and is reduced in subsequent cycles. The degree of transient behaviour is substantially less than those reported in literatures, and it could be attributed to equilibration to the sudden change of illumination condition[46–48]. Overall, very little changes in performance were observed during the approximately 550 h light cycling test. We believe that the light cycling stability is highly architecture dependent and the degradation/recovery is not critical for the device stack in this study.

UV stress study

UV stress tests were conducted to understand the impact of UV light exposure on the device stability. The standard UVA-340 lamp of 1 Wm^2 nm at 340 nm was used to continuously illuminate the devices, and the results are shown in Extended Data Fig. 7a. In this test, the devices maintained an average of 97% of initial device PCEs, with a range of 93% to greater than 99% for 6 individual devices with approximately 14 kWh m^{-2} exposure. Note that 15 kWh m^{-2} is a condition used in International Electrotechnical Commission 61215 UV exposure test[49]. The devices maintained an average of 94%, with a range of 92% to 96% with controlled UV dose of approximately 34 kWh m^{-2}. A separate UV test was conducted using continuous 4 W UV lamp (365 nm) illumination over 1,125 h, and the results are shown Extended Data Fig. 7b. In this test, the devices maintained an average of 96.4% of initial device PCEs, with a range of 94.2% to 97.9% for 12 individual devices. These UV tests show that for the devices used in this study the UV exposure has a relatively small impact on the device degradation.

HTL modification for improving interfacial stability

Two SAM materials (MeO-2PACz and Me-4PACz) were compared for evaluating their potential impact at the bottom interface using DFT calculations and electrochemical measurements. Both SAMs were previously shown to be effective HTLs for p–i–n PSCs[7], and MeO-2PACz was used as the HTL SAM in our standard device configuration[2] (Figs. 1–3). It is known that ITO reacts with methylammonium (MA)- and formamidinium (FA)-containing perovskites under thermal annealing and/or voltage bias conditions, so the small amounts of MA and FA, or even HI, released from the perovskite under illumination can be expected to react with ITO if they can penetrate the HTL SAM[35]. The DFT calculations showed that MeO-2PACz has a stronger binding energy towards H_2O (0.34 eV) than Me-4PACz (0.17 eV) (Extended Data Fig. 8a). Thus, MeO-2PACz has a higher affinity to attract or trap more H_2O/moisture than Me-4PACz, which can accelerate analogous ionic degradation reactions[50] or facilitate proton transfer across the SAM layer via H_3O^+ intermediates. We conducted DFT calculations to further examine the interaction between SAM (MeO-2PACz or Me-4PACz) and perovskite (using $FAPbI_3$ as an example), as shown in Extended Data Fig. 8b. We found that MeO-2PACz has a stronger interaction with perovskite (0.67 eV) than Me-4PACz (0.52 eV). Again, the computational results suggest that the polarity and hydrogen bonding of the MeO group on the MeO-2PACz interacts more strongly with ions than non-polar Me-4PACz, and we expect this property may increase the ability of MA or FA to reach the ITO interface through the HTL.

We subsequently used electrochemical measurements to characterize the ion-blocking properties of MeO-2PACz, Me-4PACz, and a 1:1 mixture of the two. Extended Data Fig. 8c shows the potentiostatic current density–time (J–t) plots of ITO and the indicated ITO/HTL samples biased at −1.05 V versus an Ag|AgCl reference electrode in a 0.2 M methylammonium chloride aqueous electrolyte. This potential is just above the threshold to induce acid-mediated (MA in this case) cathodic reduction of ITO to In^0 via the following reaction[51]:

$$In_2O_3 + 6MA^+ + 6e^- \rightarrow 2In^0 + 3H_2O + 6CH_3NH_2$$

The current density in Extended Data Fig. 8c correlates to the rate at which MA^+ permeates the HTL. A lower current density and slower transient indicate that Me-4PACz is better at blocking MA^+ to slow the reaction kinetics, consistent with the conclusions drawn from DFT. Similarly, we expect Me-4PACz to impede ionic reactions in a solid-state device and will improve the p–i–n PSC operational stability over MeO-2PACz as an HTL. However, because Me-4PACz is less hydrophilic than MeO-2PACz and often presents a challenge for perovskite wetting/coating, a mixture of MeO-2PACz and Me-4PACz can be used to enable both good surface wetting (Extended Data Fig. 8d,e). The mixed HTL showed improved ion-blocking ability (Extended Data Fig. 8c) and can be expected to enhance the interfacial stability as well.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request.

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Author contributions Q.J. and K.Z. conceived the idea. Q.J. fabricated perovskite solar cells and conducted device efficiency measurements. R.T. conducted device operational stability measurements at different temperatures under the supervision of J.J.B. Q.J. packaged the devices for outdoor operation, damp heat, and thermal cycling studies, with guidance and help from E.A.G. R.A.K. conducted electrochemical measurements and analysis. J.M.N. conducted the thermal cycling study and UV stress study. Y.X. and X.W. conducted the DFT calculation and analysis under the supervision of Y.Y. Q.J. and K.Z. wrote the first draft of the manuscript. All authors discussed the results and contributed to the revisions of the manuscript.

Competing interests The authors declare no competing interests.

Additional information

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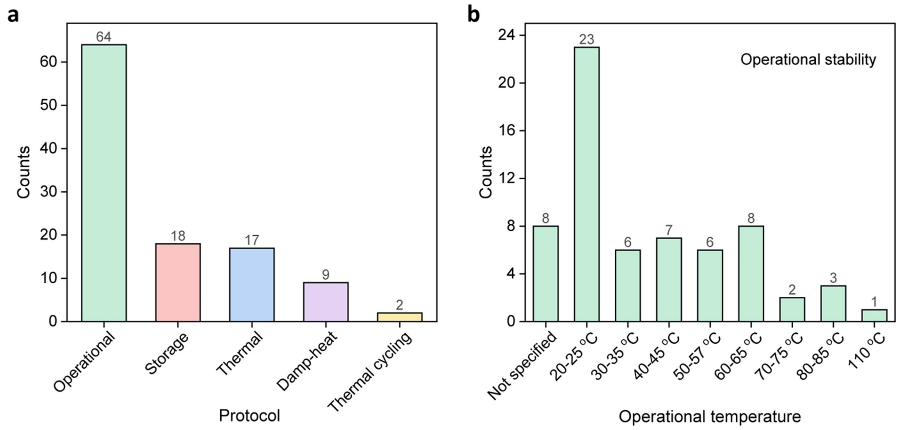
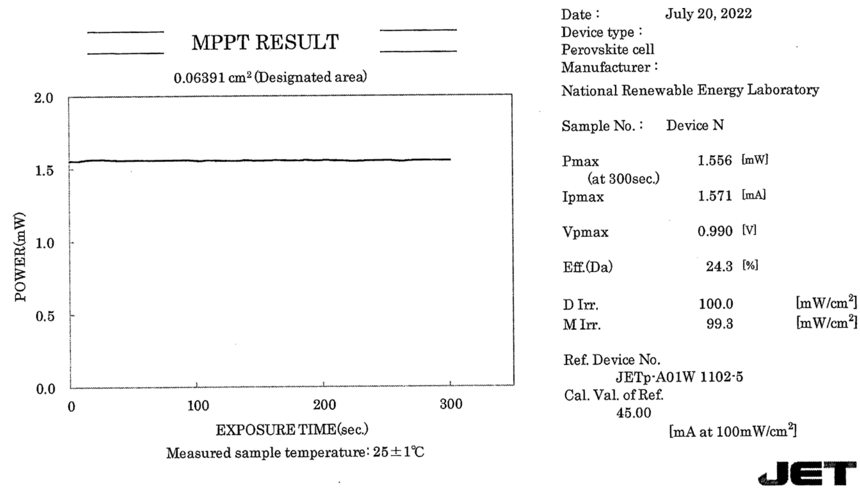
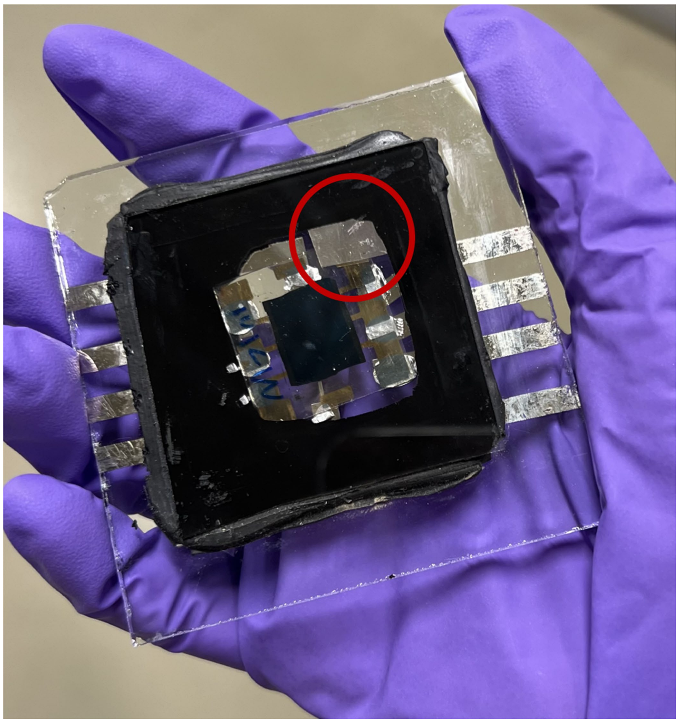


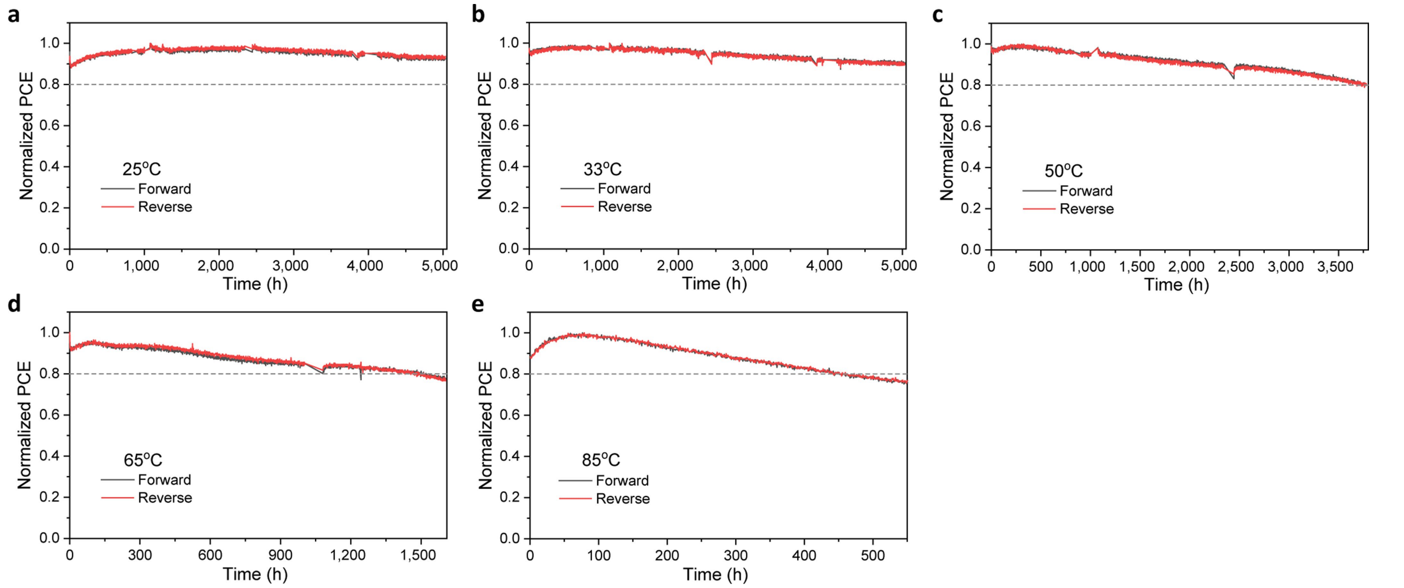
Extended Data Fig. 1 | Statistical analysis of stability testing protocols. **a**, Analysis of different protocols used in recent publications as summarized in Supplementary Table 1. Of the various ISOS stability testing protocols, the published literature clearly focuses on the operational protocol. **b**, The temperatures used for operational stability vary across a wide range, with most such tests conducted at room temperature.



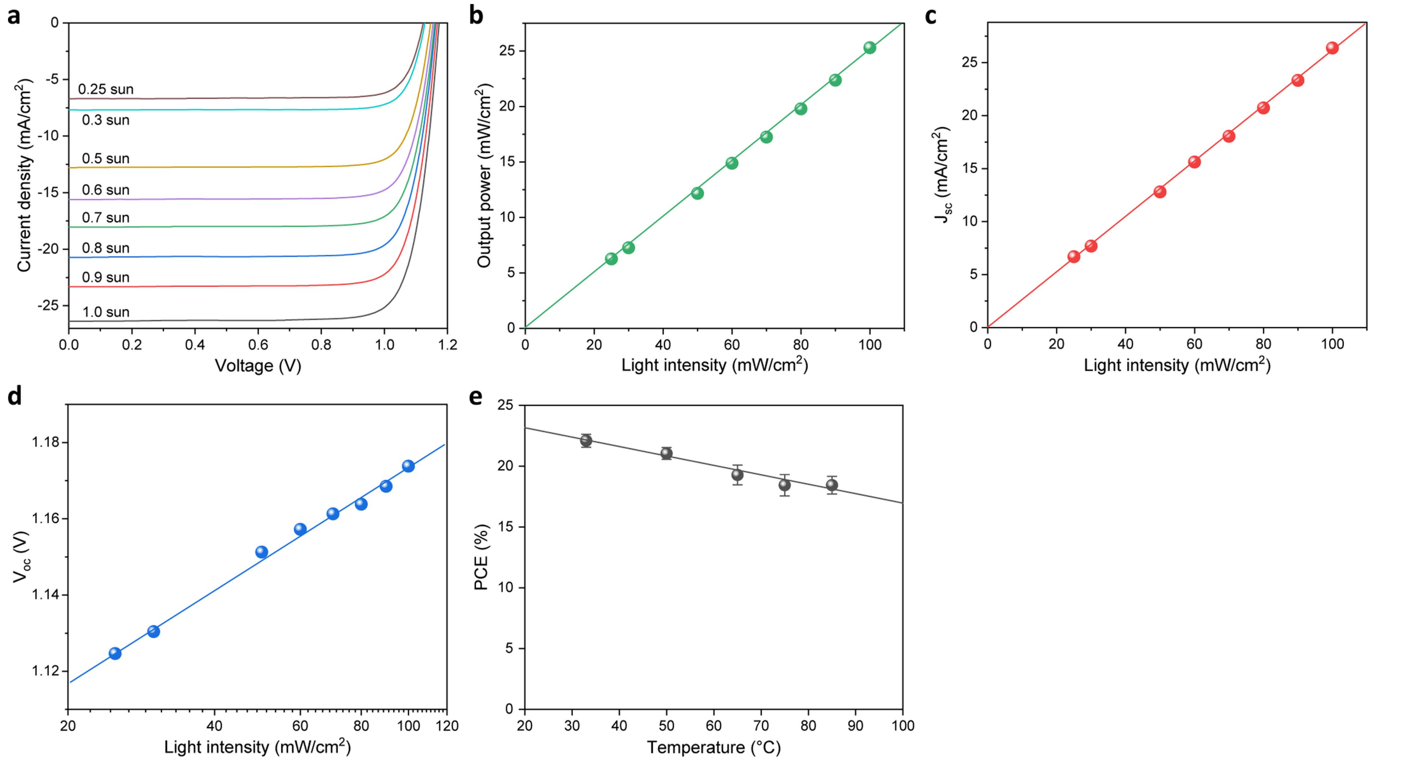
Extended Data Fig. 2 | Device performance certification. The certification results of a representative device measured by an accredited independent photovoltaic (PV) calibration and measurement laboratory (Japan Electrical Safety and Environment Technology Laboratories, JET). The device shows a stabilized power conversion efficiency (PCE) of 24.3%.



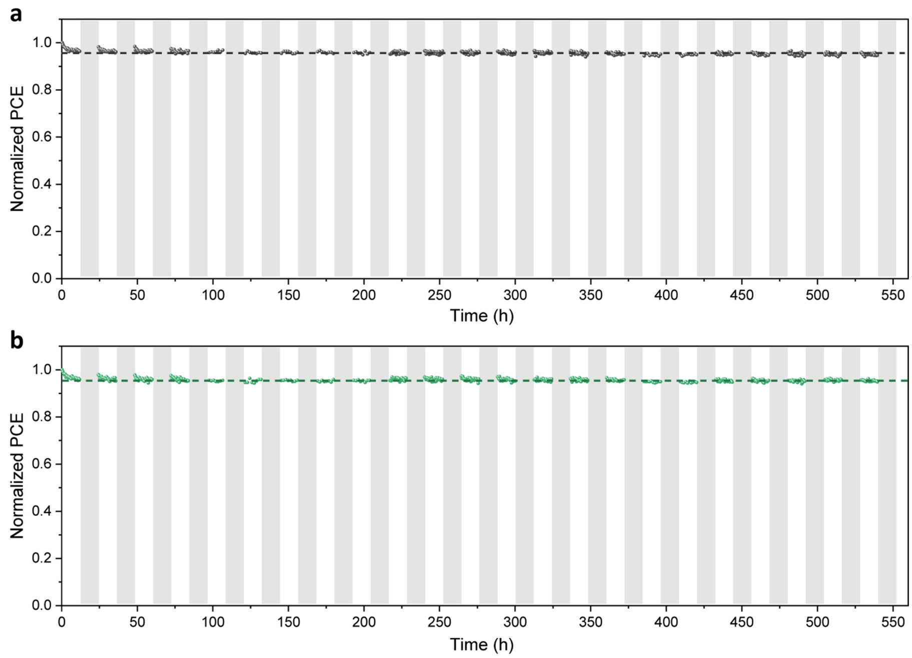
Extended Data Fig. 3 | Photograph of a representative packaged device. A calcium flake (indicated by the red circle) deposited onto PET flexible substrate was placed inside the package as a sensor to monitor moisture ingress.



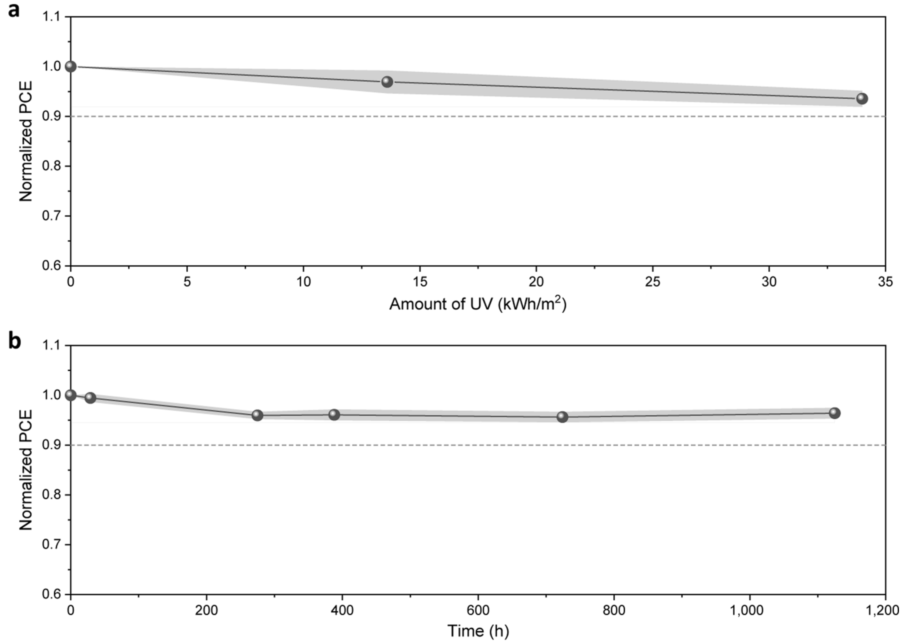
Extended Data Fig. 4 | Typical operational stability of PSCs at different temperatures. **a**–**e**, Operational stability tests with the temperature controlled at 25 °C (**a**), 33 °C (**b**), 50 °C (**c**), 65 °C (**d**), and 85 °C (**e**), respectively. Both reverse and forward J–V scan results are used as indicated.



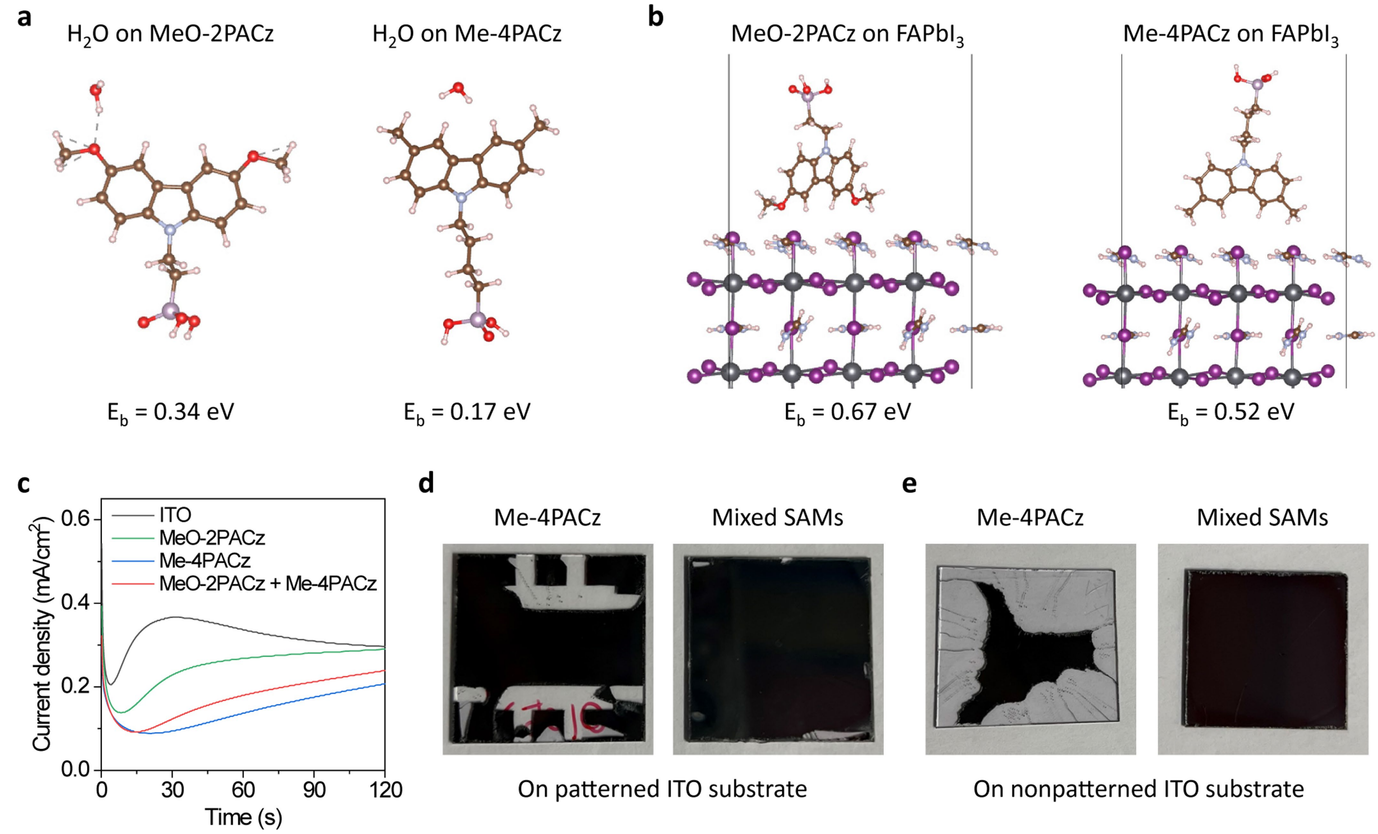
Extended Data Fig. 5 | Effect of light intensity and temperature on PSC performance. a–d, Effect of changing AM1.5 G illumination intensity (0.25 to 1.0 suns) on the J–V curves (**a**), output power (**b**), short-circuit current density J_{sc} (**c**), and open-circuit voltage (V_{oc}) (**d**) for a typical PSC. **e**, Effect of varying operational temperature on the PCE of PSCs. The temperature coefficient is $k = -0.077\%/K$.



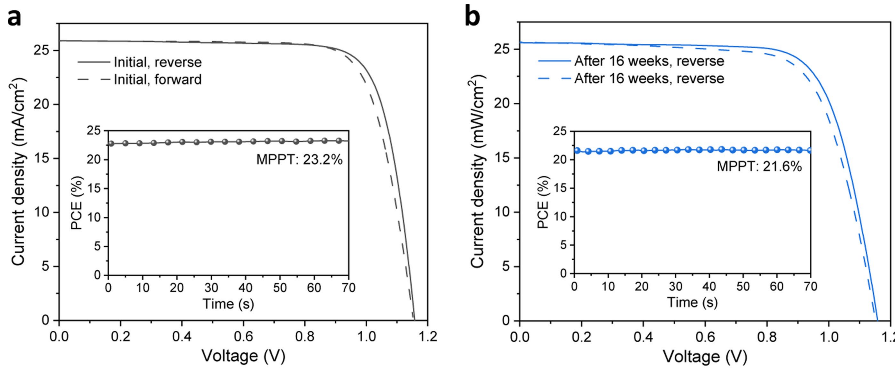
Extended Data Fig. 6 | Simulated diurnal cycle study. a, **b**, Two typical examples are shown in (**a**) and (**b**), respectively. The diurnal cycle was simulated by using repeated 12-hour light on and 12-hour light off cycles. The temperature was controlled at approximately 35 °C.



Extended Data Fig. 7 | UV stress study. a, UV exposure using the lamp UVA-340, with 1 W/m^2/nm at 340 nm. The error bars represent the standard deviations from 6 individual cells. **b**, UV exposure using a 4-W UV lamp, 365 nm wavelength. The error bars represent the standard deviations from 12 individual cells.



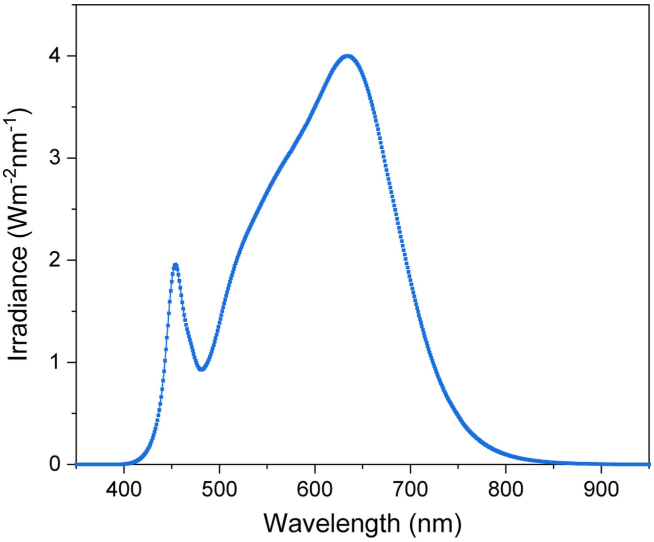
Extended Data Fig. 8 | SAM comparison. a, Configuration of H_2O molecule binding with MeO-2PACz and Me-4PACz molecules. **b**, Configuration of MeO-2PACz and Me-4PACz molecules on perovskite (100) surface. **c**, Potentiostatic current density–time (J-t) plots for the indicated ITO/HTL held at −1.05 V versus Ag|AgCl in a 0.2-M methylammonium chloride (MACl) aqueous electrolyte, just above the threshold to induce MA-mediated $In^{3+} \rightarrow In^{0}$ reduction. **d**, **e**, Photographs comparing perovskite deposition on top of Me-4PACz and mixed MeO-2PACz and Me-4PACz on patterned (**d**) and nonpatterned (**e**) ITO substrates.



c

Outdoor	Scan	J_{sc} (mA/cm²)	V_{oc} (V)	FF (%)	PCE (%)	MPPT (%)
Initial	Reverse	25.89	1.159	78.2	23.5	23.2
	Forward	25.88	1.154	77.1	23.0	
16 weeks	Reverse	25.59	1.159	73.6	21.8	21.6
	Forward	25.64	1.152	71.3	21.1	

Extended Data Fig. 9 | J–V measurement and maximum power point tracking (MPPT) of a typical packaged PSC. The measurements were conducted prior to outdoor test and after 16 weeks of outdoor aging.



Extended Data Fig. 10 | Typical spectrum of white LED-based solar simulator. This lamp was used for indoor, temperature-dependent stability tests.